

Peer Review Filter Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	peer-review-filter

Big idea: Scientific knowledge is filtered by community criticism, not just produced by individual discovery.

Student-Facing Prompt

Inspect Peer Review Filter Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Peer Review Filter Stimulus A	Student-facing stimulus 1: Study claim: a lavender scent improves memory scores by 18% in one class of 18 students. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Peer Review Filter Stimulus B	Student-facing stimulus 2: Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Peer Review Filter Reveal / Twist	Student-facing stimulus 3: Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Scientific knowledge is filtered by community criticism, not just produced by individual discovery.

Actual Lesson Questions

#	Question for students
1	After inspecting Peer Review Filter Stimulus A, what is your first knowledge claim?
2	Which exact detail from Peer Review Filter Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	How does criticism strengthen knowledge?

Ready-to-Use Examples

- Study claim: a lavender scent improves memory scores by 18% in one class of 18 students.
- Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Give groups a fictional study with several strengths and weaknesses.
2. Students annotate method, sample, measurement, claim size, and limitations.
3. Groups write reviewer comments and a decision: accept, revise, or reject.
4. Compare reviewer decisions and identify criteria behind disagreement.
5. Debrief peer review as a social process for improving knowledge.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: How does criticism strengthen knowledge?
- Does peer review guarantee truth or reduce risk?
- What should a scientific community do with promising but flawed evidence?

Knowledge Questions

- How does criticism strengthen knowledge?
- What role do communities play in scientific reliability?
- When is evidence strong enough to publish?

Transfer Example

For example, run this activity with a mock paper evaluated by reviewers with different priorities. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...